FINAL EXAMINATION OF SHANGHAI JIAO TONG UNIVERSITY (A)

(2020 – 2021 First Semester)

Class: __________________ Student No.: __________________ Name: __________________

Course: Discrete Mathematics (MA208) Score: ________

Problem 1. (**Logic, 12'**) Suppose P, Q and R are propositional variables. Find a proposition in CNF (conjunctive normal form) which is logically equivalent with $(P \wedge Q) \leftrightarrow R$, and explain why they are logically equivalent.

Problem 2. (**Logic, 12'**)

- (**For Hongfei Fu's class.**) Suppose P, Q and R are predicate symbols, and the domains of the variables x, y are the same. Prove that

$$\forall x(P(x) \to (Q(x) \to R(x))) \equiv \forall y((P(y) \wedge Q(y)) \to R(y)).$$

- (**For Qinxiang Cao's class.**) Suppose P, Q and R are predicate symbols. Prove that

$$\forall x(P(x) \to (Q(x) \to R(x))) \equiv \forall y((P(y) \wedge Q(y)) \to R(y)).$$

I pledge that I will obey examination disciplines. Signature:

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Number	1	2	3	4	5	6	7	8
Score								
Signature								

Problem 3. (**Set Theory, 12'**) Suppose $\bigcup X$ represents $\{y \mid \exists x(x \in X \wedge y \in x)\}$ for any set X. Prove that if $A \subseteq B$, then $\bigcup(\bigcup(\bigcup A)) \subseteq \bigcup(\bigcup(\bigcup B))$.

Problem 4. (**Set Theory, 16'**) Suppose R^+ represents the transitive closure of R.
(**Note:** in the textbook we use R^* for the transitive closure of R.)

- If a relation $R \subseteq A \times A$ on a set A is reflexive, is it always true that $(R \cap R^{-1})^+$ is an equivalence relation on A? Please either prove or disprove your answer. (**8'**)
- If a relation $R \subseteq A \times A$ on a set A is **not** reflexive, is it always true that $(R \cap R^{-1})^+$ is **not** an equivalence relation on A? Please either prove or disprove your answer. (**8'**)

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Problem 5. (**Graph Theory, 12'**) Consider 7 letters A, B, C, D, E, F, G with frequencies 0.16 0.08 0.30 0.09 0.17 0.11 0.09, respectively. Please use Huffman coding to encode these symbols so that the average number of bits required to encode a symbol is minimum. **(You need to demonstrate all intermediate results.)**

Problem 6. (**Graph Theory, 12'**) Suppose G_1 and G_2 are any two simple graphs, homeomorphic to each other. Prove or disprove the following statement: G_1 has a Hamilton path if and only if G_2 has a Hamilton path.

Problem 7. (**Graph Theory, 12'**) Suppose $G = (V, E)$ a weighted connected simple graph with weight function $w : E \to \mathbb{R}$, and $e_0 \in E$ has the greatest weight, i.e. for any $e \in E \setminus \{e_0\}$, $w(e_0) > w(e)$. Prove that if e_0 is in a minimum spanning tree of G, then e_0 must be a **cut edge** of G.
Hint: For connected simple graph $G = (V, E)$, an edge e of G is a **cut edge** iff $G - \{e\}(= (V, E \setminus \{e\}))$ is disconnected.

Problem 8. (**The Inclusion-Exclusion Principle, 12'**) For any natural number $n \geq 1$, prove that

$$n! = \sum_{k=0}^{n} (-1)^{n-k} C_n^k k^n.$$

Hint 1: consider how to calculate the number of all bijections from $\{1, 2, 3, \ldots, n\}$ to $\{1, 2, 3, \ldots, n\}$.
Hint 2: k^n is the number of functions from $\{1, 2, 3, \ldots, n\}$ to $\{x_1, x_2, \ldots, x_k\}$ where $x_1 < x_2 < \cdots < x_k$.